ElimuSentry - Phase 2: Pipeline Execution and Model Diagnostics

What was built, what the diagnostics showed, and where reality changed the plan

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04-652 Artificial Intelligence System Design, Assignment 2, Phase 2 (Strategy Execution, Pipeline and Visual Diagnostics)
Repository: https://github.com/Emile-Lucky-Muhigira/AISD_Project

***Abstract*—Phase 1 planned, with evidence, how every piece of information about a child in the Ghana Socioeconomic Panel Survey would be prepared for a school-dropout early-warning model. This report covers Phase 2: turning that plan into working code and checking it. The plan was built as one scikit-learn `Pipeline` around a `ColumnTransformer`, with custom preparation steps in `src/transformers.py`. It is fitted on the model-building children only (X_train, 1,307 children from 2009/10) and then applied unchanged to the testing children (X_test, 1,395 children from 2013/14). The output has exactly the planned size: 33 features become 30, and then 53 numeric columns for XGBoost or 48 for logistic regression, with no missing value left. The three removed features can be rebuilt exactly from the kept ones (0 differences in 3,921 checks), so the structural removals add no extra loss beyond the retained source fields. Eight diagnostic figures inspect the preprocessing steps. Most decisions were carried out as planned; five points changed or were sharpened by what the data showed. A retained fixed CPI adjustment aligns nominal monetary units; other cross-period changes remain descriptive. MI-only feature selection is unstable across folds; a fixed structural/domain rule determines the removals. The hours-missed rule is compared descriptively with median filling; no equivalence claim is made. All four "was missing" flags are less frequent later, by different proportions; survey mode alone does not establish the cause. And rank correlation was needed because the ordinary correlation would have missed one repeating pair. The professor-requested extension evaluates six model families with nested district-grouped tuning and training-only probability calibration. Extra Trees, selected from training validation, achieves average precision 0.134 on the primary later cohort and catches 20 of 76 dropouts with 126 false alarms at 20% capacity. Explanations and drift diagnostics identify strong geographic dependence and changing missingness patterns. Successful preprocessing execution therefore does not demonstrate reliable temporal prediction.**

***Index Terms*—data preprocessing, scikit-learn pipeline, leakage prevention, visual diagnostics, school dropout, Ghana panel survey**

I. Introduction

Aim of this phase. Phase 1 used the Ghana Socioeconomic Panel Survey [2] and decided how to fill gaps, treat extreme values, remove repeating features, encode categories, rescale numbers and split the data with fold-fitted preprocessing [1]. Phase 2 has three tasks, which follow the assignment: (1) implement that strategy as fully runnable code with a strict X_train / X_test separation; (2) check every assumption with eight diagnostic figures; and (3) record where reality agreed with the plan and where it forced a change.

What is in the repository. `src/transformers.py` holds the custom preparation steps; `src/pipeline_builder.py` assembles them into the pipeline; `phase2_implementation/notebooks/pipeline_execution.ipynb` runs everything from start to finish, saves the eight figures to `figures/` at 300 DPI, writes the processed tables to `data/processed/` and the decision log to `phase2_implementation/decision_log.csv`. The execution verification at the end of this report records the status of the integrated notebook rerun.

Scope. The original eight preprocessing diagnostics are retained. Professor feedback extends the analysis to six model families, training-only tuning and calibration, intervention policies, subgroup performance, retrospective temporal evaluation, and model explanations. Later-wave outcomes are used to report performance, never to fit the frozen model, calibrator, or threshold. Repeated training-CV experiments and earlier descriptive inspection of later outcomes are disclosed: this is not untouched external confirmation.

Reading this report. The opening sections preserve the pipeline, eight visual diagnostics, decision log, and output checks. The model-evaluation section adds the professor-requested evidence and separates exploratory alternatives from the frozen evaluation. The final sections describe limitations and verification; the Appendix explains terminology.

II. What Was Built

One pipeline, two versions. The pipeline is built with scikit-learn [4]. The same steps produce two versions of the data (Table I). The **XGBoost version** [5] keeps numbers in their own units, because a decision tree only compares values; it has 53 columns. The **logistic regression and PCA version** adds the log on long-tailed features and standardises the 16 numeric/ordinal columns, retains 32 indicator columns on their 0/1 scales, and leaves out one option per label as the reference; it has 48 columns. This follows Phase 1, C1 exactly.

How leakage is prevented. Every step that learns something (averages, middle values, neighbours, cap limits, money ranks, the list of options, the scaler) is inside the pipeline. The pipeline is *fitted* on X_train only and then only *applied* to X_test. Inside cross-validation it is fitted again on the building part of each fold. The notebook checks this directly: it saves the

TABLE I
THE PIPELINE, STEP BY STEP, IN THE ORDER THE DATA PASS THROUGH IT

#	Step (code name)	What it does	Learns from	Phase 1
0	`deflate_to_2009_prices`	Divides 2013/14 money amounts by 1.62 to express them in 2009/10 prices	Nothing (fixed number)	C1
1	`MissedHoursRule`	A blank "hours missed" becomes 0 when attendance was recorded	Nothing (fixed rule)	B1
2	`MissingFlagger`	Adds the four yes/no "was missing" flags before any gap is filled	Nothing (fixed rule)	B1
3	`GradeKNNImputer`	Mixed-distance grade KNN; k selected in inner district-grouped training folds; recalculates over-age	X_train (donors, numeric ranges and inner-selected k)	B1
4	`ColumnDropper`	Removes people per room, lost a parent and school grade (33 → 30)	Nothing (fixed list)	B4b
5	`CategoryRecoder`	Turns 1/2 answers into 0/1, reverses the textbooks scale, groups relationship to head into 3 options	Nothing (fixed rule)	C1
6	`SimpleImputer` (in the `ColumnTransformer`)	Fills gaps: average for balanced numbers, middle value for lopsided ones, most common answer for yes/no and ordered items, a "missing" option for school type	X_train (averages, middle values, most common answers)	B1
7	`LogIQRCapper`	Caps travel time, household size and rooms at the IQR limit on the log scale; adds the log for the simple model	X_train (limits)	B2, B3
8	`TrainingRank`	Turns the two money amounts into a 0–1 rank against the model-building children	X_train (reference values)	C1
9	`OneHotEncoder`	One yes/no column per option of each label	X_train (list of options)	C1
10	`StandardScaler` (simple model only)	Standardises 16 numeric/ordinal columns (average 0, spread 1); 0/1 indicators remain unscaled	X_train (average, spread)	C1

Steps 1–5 run one after another on the whole table because each one needs the result of the one before (for example, flags must be added before gaps are filled). Steps 6–10 run side by side on different groups of columns inside the `ColumnTransformer`. Step 0 is applied to X_test before the pipeline. Code: `src/transformers.py` and `src/pipeline_builder.py`.

fitted pipeline, applies it to X_test, and confirms that nothing it learned has changed [6]. The only step applied outside the pipeline, the inflation adjustment, uses a fixed published number, not the data.

The split. Inputs contain 1,307 training and 1,395 later-period children. Primary temporal evaluation is restricted to the 731 children in 605 new households (76 dropouts), with no shared household or person. Secondary evaluation uses 1,175 new people but allows 370 shared households. Five training folds keep whole districts together. Aggregate later-period covariates and outcomes were already explored, so this is retrospective temporal evaluation, not an untouched external confirmation. These subsets are fixed by IDs, not their outcomes.

The inflation factor. Phase 1 promised to convert 2013/14 money to 2009/10 prices using Ghana's consumer price index. The interview dates in the raw survey files show that the first visit took place from November 2009 to April 2010 and the second, labelled 2013/14, mostly from March to October 2014, with a few households in 2015. Using World Bank annual consumer price inflation for Ghana [3] (index 2009 = 100: 2010 = 110.7, 2014 = 172.6, 2015 = 202.3), weighted by the number of interviews in each year, the price level was 107.5 at the first visit and 174.0 at the second. Every 2013/14 amount is therefore divided by $174.0/107.5 = 1.62$.

III. VISUAL DIAGNOSTICS

Missing answers (Fig. 1.1). The pipeline sees the same pattern as Phase 1: 3,847 of 43,131 feature cells missing in X_train (8.92%) and 1,827 of 46,035 in X_test (3.97%); 53 complete children in X_train and 467 in X_test. After the pipeline, **no missing value is left** in either table.

Outcome (Fig. 1.2). Dropouts are the small group in both periods, so each dropout is given extra weight during training (`scale_pos_weight` for XGBoost, `class_weight="balanced"` for logistic regression), recalculated inside every fold.

Extreme values (Fig. 2.1). The limits learned by the pipeline (IQR rule on the log scale [8]) are exactly those planned (294.0 minutes, 15.2 people, 18.8 rooms), and exactly 13 values are capped in X_train (2 travel times, 6 household sizes, 5 room counts) and 5 in X_test. The two money features are no longer shown: as planned in Phase 1, C1, they are ranked instead of capped, so their extreme values cannot dominate.

Repeating features (Figs. 3.1 and 3.2). The same two pairs are flagged as in Phase 1: father alive and lost a parent ($\rho = -0.850$), rooms and people per room ($\rho = -0.855$). The third case, school grade, is an exact calculation from age and years over-age (panel c). The fitted lines confirm that each removed feature follows directly from features that are kept.

Links with dropout (Fig. 3.3). The ranking (mutual information [9]) is the one reported in Phase 1: region first (mutual information 0.037), then household goods, rooms, school spending, English score, household size and parents' education. No feature is removed for a weak link: any data-driven feature choice must remain inside training folds.

Transformations (Fig. 4.1). The log is the step that changes the shape; Min–Max and standardising only change units. This confirms the Phase 1 choice: log on long tails, then `StandardScaler` for the simple model and PCA, and no rescaling for XGBoost.

Dimension reduction (Fig. 5.1). The first component captures 16.6% of the variation, and the curve then rises slowly: 21 of 48 components are needed for 90% and 28 for 95%. These variance shares describe the current mixed-scaled representation, not an analysis in which all 48 columns have unit variance. PCA could reduce width, but would replace

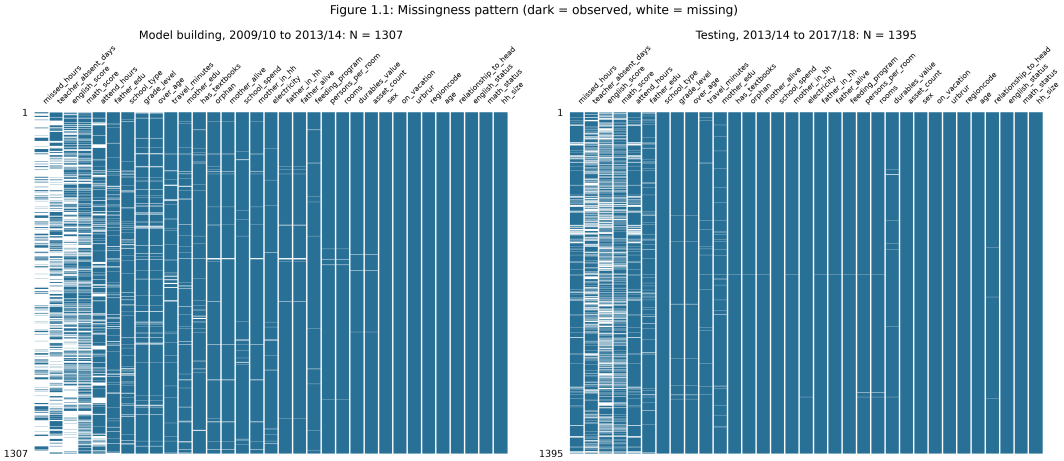


Fig. 1.1. Where answers are missing. Left: X_train; right: X_test. *How to read it:* each thin row is one child and each column one feature; white marks are missing answers. The large white blocks on the left of X_train are hours missed, days teacher absent, the two test scores and hours attended. X_test has fewer gaps; survey routing, recording and cohort composition may contribute, but the figure does not identify the cause.

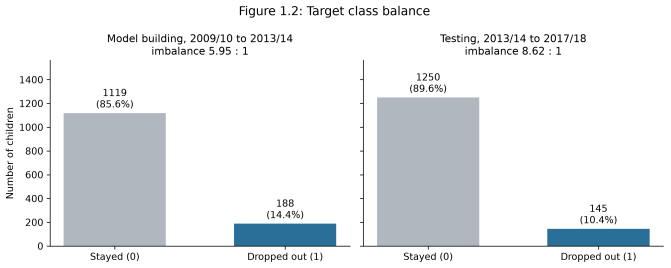


Fig. 1.2. Children who stayed in school and who dropped out. X_train: 188 of 1,307 dropped out (14.4%, imbalance 5.95 : 1); X_test: 145 of 1,395 (10.4%, 8.62 : 1).

named features with combined components that are harder to explain, which matters for an early-warning system whose warnings must be explained [12]. PCA is therefore kept as a diagnostic and not used to replace the features [7].

IV. Decision Log: Plan vs Implementation

Tables II and III distinguish the original preprocessing plan from the later recommendations. Calibration and retrospective temporal scoring are complete. Alternative cleaning and prediction policies remain separate experiments; their apparent gains do not replace the frozen temporal result.

V. Deviations and Rationale

A. *Price adjustment and rank interpretation*

Retain the fixed external CPI conversion (factor 1.62) and learn the rank reference on training rows only. Training-reference ranks bound long-tailed monetary inputs without fitting the later distribution. Previously explored later-period ranks differ from the training median, but that alone cannot identify a causal price or wealth change: composition, measurement and attrition can contribute. No claim is made that inflation explains most of the change or that school policy caused the spending difference. Within-wave test ranking is not used to fit the primary procedure or to choose it from test scores.

B. *Which repeating feature to drop is not stable across folds*

The removal list is fixed from feature construction and domain meaning, not from full-training dropout associations. Retain household size and rooms instead of derived crowding; retain parent-alive source fields instead of derived orphan status; retain age and the domain-relevant over-age signal instead of grade. The separate MI-only diagnostic agrees in 1 of 5 folds (Table IV), so it is not represented as a stable selection algorithm. This diagnostic does not change the fixed list. Source identities are checked on all training records, including matching missing values; this is a check of structural removal, not a claim that all preprocessing is lossless.

C. *Hours missed: an assumption with a sensitivity comparison*

The retained rule assumes zero missed hours when attendance is recorded; it is not recovery of a known survey answer. It sets 756 blanks to zero and leaves 261 holiday/unknown-attendance cases for the later median fill. The untuned five-fold comparison gives logistic-regression PR-AUC 0.266 for the rule versus 0.275 for median filling, and XGBoost 0.252 versus 0.257. These are descriptive scores with substantial fold variation; no formal equivalence or no-harm conclusion is supported. The sensitivity remains separate; the later source-audited cleaning comparison revisits this assumption under grouped training validation. Temporal test scores must not select it.

D. *Missing indicators can change meaning across periods*

Flags are on less often in the later cohort: teacher absent 56.8% vs 31.8%, hours attended 21.8% vs 11.7%, father's education 7.7% vs 5.4%, mother's education 3.8% vs 1.9%. Survey mode, coding and cohort composition may all matter; tablet use alone is not established as the cause. Retain the four design/family-context flags as a fixed modelling choice, not

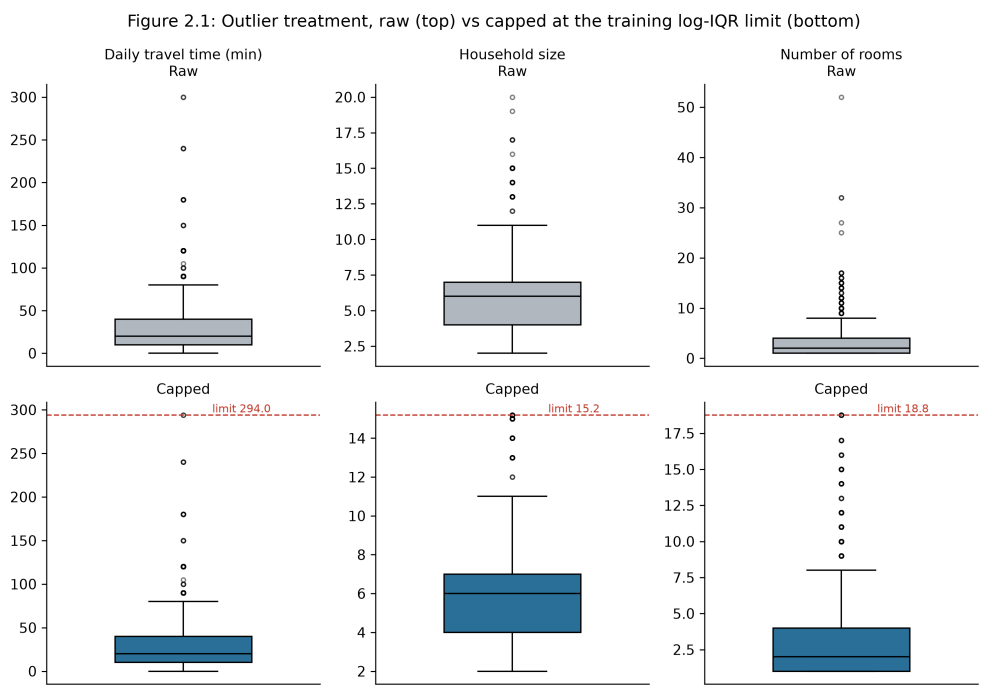


Fig. 2.1. Extreme values before (top) and after (bottom) capping, for the three capped features. *How to read it:* the box holds the middle half of the children and the line inside it is the middle value; dots are unusually high values. The red dashed line is the limit learned from X_train: 294.0 minutes, 15.2 people and 18.8 rooms. Values above the line are pulled down to it; no child is deleted.

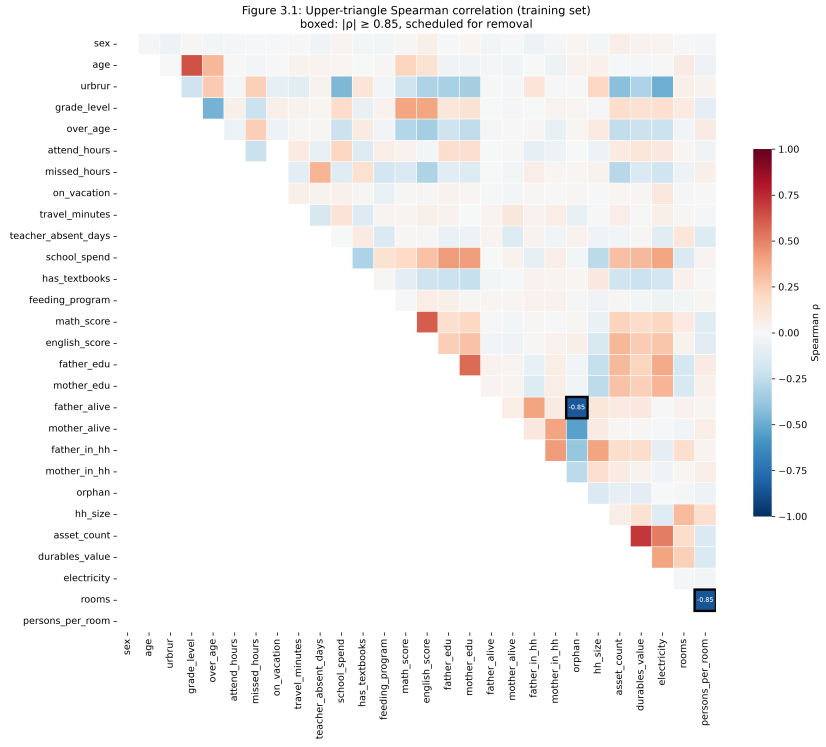


Fig. 3.1. Which features repeat each other (rank correlation, X_train). *How to read it:* each square compares two features; dark red means they rise together, dark blue that one falls as the other rises. The two boxed squares are the pairs at or above 0.85 (ignoring the sign), which are scheduled for removal.

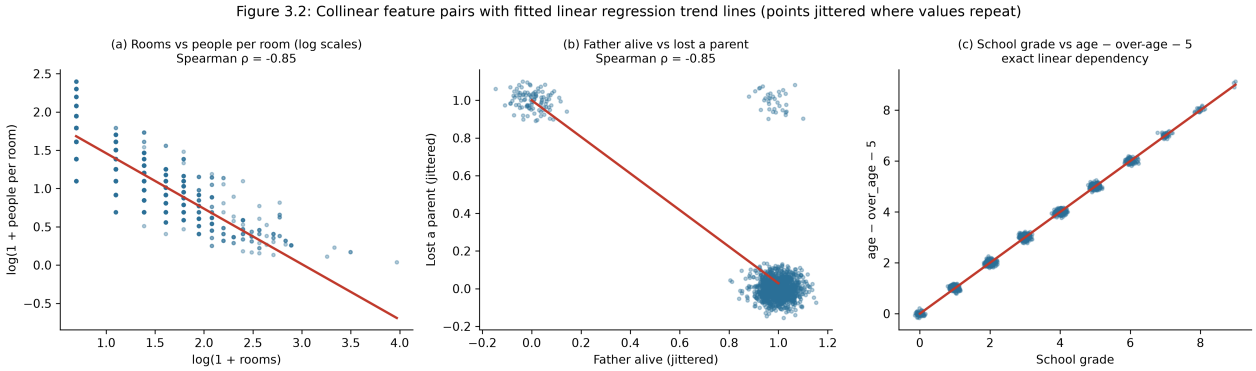


Fig. 3.2. The three repeating features, one dot per child, with a fitted straight line. (a) More rooms, fewer people per room (log scales). (b) "Lost a parent" is almost the opposite of "father alive" (dots spread slightly so they do not sit on top of each other). (c) School grade equals age − years over-age − 5 for every child, so every dot lies on the line.

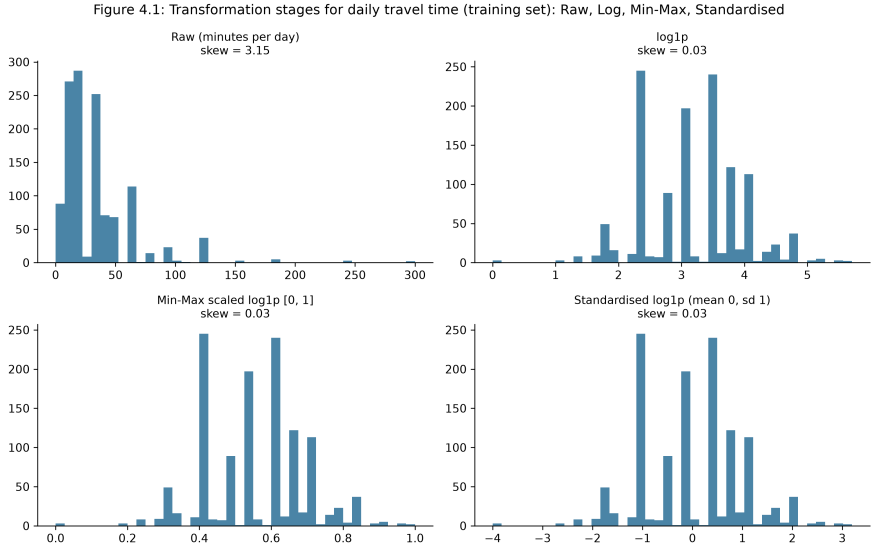


Fig. 4.1. What the transformation stages do to daily travel time (X_train). Top left: raw minutes, with a long tail to the right (skew 3.15). Top right: after the log, the shape is balanced (skew 0.03). Bottom: the same logged values squeezed into 0–1 (Min–Max) and standardised (average 0, spread 1); the shape does not change, only the units on the bottom axis.

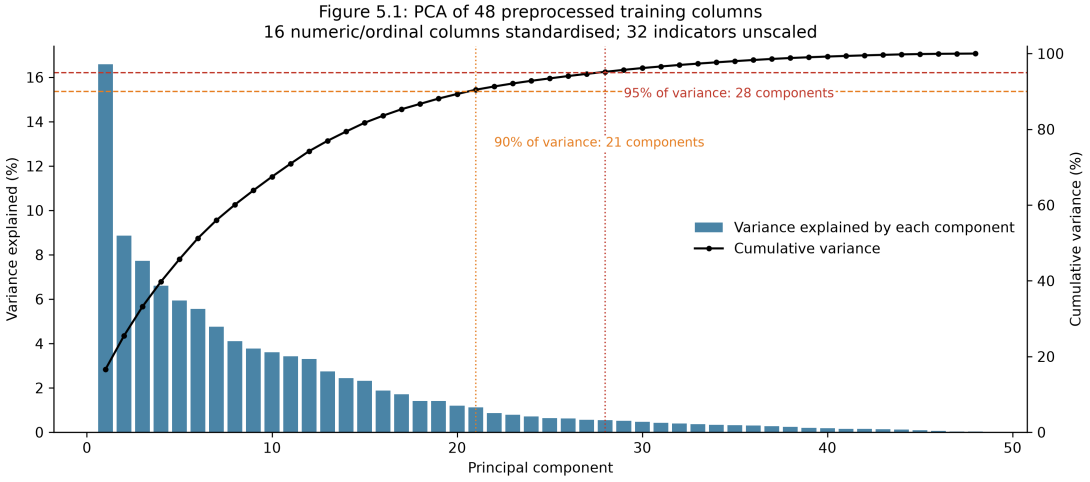


Fig. 5.1. Principal component analysis of 48 preprocessed training columns: 16 numeric/ordinal columns standardised and 32 indicators left unscaled. PCA centres these inputs without further scaling. *How to read it:* each bar is the share of the total variation captured by one combined component; the black line adds them up. The dashed lines mark 90% and 95% of the variation, reached with 21 and 28 components.

TABLE II
ORIGINAL PREPROCESSING DECISIONS: PLAN VERSUS EXECUTION

Step	Plan or recommendation	Implementation and evidence	Status
Fix recording errors	Fixed rules (codes, impossible values)	Applied in 01_build_dataset; nothing learned	As planned
Inflation adjustment	Convert 2013/14 money to 2009/10 prices (CPI)	Factor 1.62 (World Bank CPI, weighted by interview months)	Fixed external price adjustment; other distribution changes remain
Split	By time; primary new households; secondary new people	X_train 1307, all temporal 1395, primary 731, secondary 1175	As planned
Hours missed	Blank = 0 when attendance recorded; median otherwise	Rule applied; CV PR-AUC (LR) 0.266 vs 0.275 with median fill (descriptive comparison; uncertainty retained)	Sensitivity comparison; no formal equivalence claim
Was-missing flags	4 flags before filling	teacher absent, hours attended, father's and mother's education	As planned
Was-missing flags across periods	Compare flags between periods	All four flags are less frequent later; relative frequencies differ by feature (see notebook missingness diagnostics); survey mode alone does not identify the cause	Confirmed limitation
Filling gaps	Mean / median / mode by skew; 'missing' option for school type	SimpleImputer fitted on X_train	As planned
School grade	Mixed-distance KNN with training-only k selection	Inner-grouped k=15 on full training; over-age recalculated	Revised and validated
Repeating features	Spearman |rho| >= 0.85, VIF > 10; drop 3; re-check in each fold	MI-only diagnostic matched the structural list in 1/5 folds; removal is fixed from source identities/domain meaning, not MI; rebuilt exactly including unknowns	Fixed structural list; MI-only check is descriptive
Extreme values	Log-IQR cap on travel, household size, rooms	Limits 294.0, 15.2, 18.8; 13 capped (train)	As planned
Money	Rank 0-1 against model-building values	TrainingRank fitted on X_train	As planned
Encoding	One-hot, ordinal, 0/1	53 columns (XGBoost), 48 (LR)	As planned
Scaling	None for XGBoost; log + StandardScaler for LR and PCA	PCA: 21 components for 90%, 28 for 95%	As planned
Cross-validation	5 folds, stratified, grouped by district	StratifiedGroupKFold; 0 shared districts or households	As planned
Class imbalance	Class weighting, recalculated per fold	scale_pos_weight / class_weight='balanced'	As planned

The combined log is reproduced by the final notebook cell in `phase2_implementation/decision_log.csv`.

because an unadjusted significance test selected them. Compare their behaviour descriptively across periods and retain the uncertainty in transport claims.

E. Rank correlation was needed, not the ordinary one

The assignment's screening rule is written as $|r| \geq 0.85$. To capture monotone curved relationships and reduce sensitivity to extreme values, Phase 1 used the rank (Spearman) correlation ρ with the same threshold. Phase 2 shows why this mattered: for rooms and people per room, $\rho = -0.855$ but the ordinary (Pearson) correlation is only $r = -0.563$, because the relationship is curved (people per room = household size / rooms). With Pearson r, this repeating pair would have been missed, and the VIF check also misses it (VIF 1.77). For father alive and lost a parent, both measures give -0.850.

VI. PIPELINE OUTPUT VERIFICATION

Final size. Table V confirms that the pipeline produces exactly the size planned in Phase 1: 33 features become 30, then 53 columns for XGBoost and 48 for logistic regression, for every child of X_train and X_test, with no missing value and no child deleted.

Feature count reduced without information loss. The assignment asks for proof that the reduction from 33 to 30 loses no information. Table VI rebuilds each removed feature from features that are kept and compares the result with the original value for every child. The rebuilt value is identical in all 3,921 checks. This establishes the structural identities before subsequent value transformations. Imputation, capping and ranking still intentionally change information; the full pipeline is not claimed to be invertible.

No leakage. (1) Every learned setting is fitted on X_train only; the notebook confirms that applying the pipeline to X_test leaves its fitted settings unchanged. (2) In cross-validation the whole pipeline is re-fitted on the building part of each fold; no district or household is shared between building and checking parts. (3) Later outcomes are not used for fitting or selection. Prior descriptive inspection is disclosed; primary households and people are disjoint.

The pipeline works end to end. Table VII shows that both untuned models, trained through the full pipeline fold by fold, show exploratory training discrimination above the no-skill reference (PR-AUC [11], about 0.25–0.27 against 0.144 for no skill). The numbers are modest, as expected before tuning and with folds made of whole districts, which stops a model from relying on local patterns it has already seen. The processed tables are saved as `data/processed/train_processed_xgb.csv` and `test_processed_xgb.csv` (53 columns plus identifiers and the outcome) and the matching `_lr` files (48 columns); the test files include

TABLE III
PROFESSOR-REQUESTED EXTENSIONS AND SUBSEQUENT EXPLORATORY WORK

Step	Plan or recommendation	Implementation and evidence	Status
Model families	Professor extension: compare six families with XGBoost benchmark	XGBoost, LightGBM, CatBoost, ElasticNet/L1, Random Forest and Extra Trees; identical district folds	Completed; no post-test model replacement
Nested tuning and calibration	Professor extension: training-only selection and OOF sigmoid/isotonic	Inner grouped searches; preprocessing refitted within fits; calibration uses training OOF scores	Completed; later outcomes excluded
Frozen temporal evaluation	Professor extension: temporal and regional assessment	Extra Trees selected by training CV; primary 731 children/76 events; 20 TP, 126 FP, 56 FN at 20% capacity; secondary 1175/115	Completed retrospectively; weak transport performance
Attrition	Professor extension: compare observed and unobserved outcomes	Baseline comparisons by follow-up and parent co-residence; unknown outcomes remain unknown	Completed descriptive audit and separate weighted sensitivity
Geography and missingness	Professor extension: ablate flags and geography	Internal ablations, calibrated corrected-data comparisons, and retrospective no-geography evaluation	Completed; sensitivity is not causal evidence
Subgroup performance	Professor extension: regional and household comparisons	Counts, false alarms, misses, calibration and district-bootstrap intervals; sparse cells flagged	Completed; allocation disparities remain
Intervention policy	Professor extension: capacity and interpretable comparators	Top-20% policy; capacity sensitivity; demographic and random comparators; high-recall and F1/F2 alternatives	Completed; no policy meets all desired error goals
Interpretability	Professor extension: global/local SHAP and logistic odds ratios	Held-out XGBoost SHAP and penalized logistic score odds ratios; frozen Extra Trees exact TreeSHAP	Completed; score associations, not causal effects
Source-audited cleaning	Additional audit requested after temporal failure	Separate vacation-code, duplicate-score and asset-total corrections; teacher-routing/attendance sensitivities; six-family training comparison	Exploratory; original tables and frozen evaluation retained
Feature enrichment	Explore baseline-time predictors within survey evidence	Separate literacy, funding and teacher-routing candidates with grouped training validation	Exploratory; no automatic adoption
Temporal drift and calibration	Professor extension: diagnose cross-wave changes	Region and missingness dominate Extra Trees; region 8 prevalence 28.0% to 12.8%, mean predicted risk 21.0%	Completed; no later-wave recalibration

The combined log is reproduced by the final notebook cell in `phase2_implementation/decision_log.csv`.

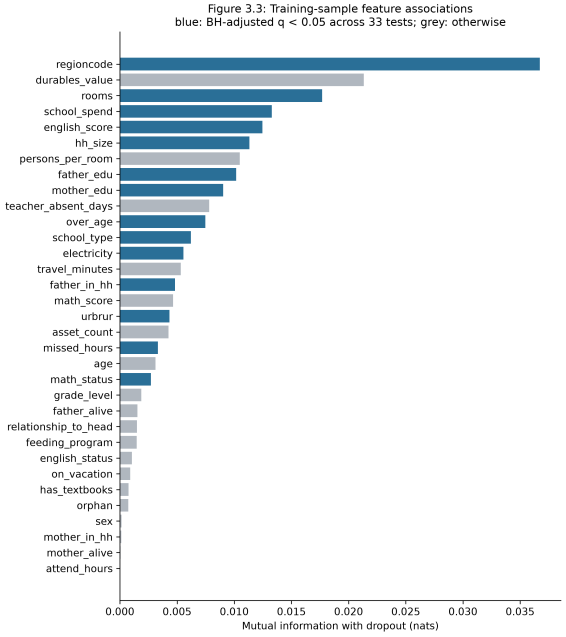


Fig. 3.3. Which features tell us most about dropout (mutual information, X_train). *How to read it:* the longer the bar, the more the feature helps to tell dropouts from children who stay. Blue: BH-adjusted $q < 0.05$ across 33 training-sample tests; grey: otherwise. Sparse tables use Fisher methods; these are exploratory associations.

TABLE IV
RE-CHECK OF THE REPEATING FEATURES INSIDE EACH FOLD

Fold	ρ rooms / people	ρ father / lost parent	Less info, pair 1	Less info, pair 2	Least info, group 3
1	-0.851	-0.842	people/room	lost parent	grade
2	-0.861	-0.831	rooms	lost parent	grade
3	-0.854	-0.861	rooms	lost parent	grade
4	-0.858	-0.854	people/room	father alive	age
5	-0.850	-0.866	rooms	lost parent	age

Computed on the building part of each fold only. Group 3 = age, grade and years over-age. Full table: section 7 of the notebook.

TABLE V
SIZE OF THE DATA AT EACH STAGE

Stage	X_train	X_test
Children (rows)	1,307	1,395
Features before the pipeline	33	33
Features after removing 3 repeating ones	30	30
Columns, XGBoost version	53	53
Columns, logistic regression / PCA version	48	48
Missing values after the pipeline	0	0
Children deleted	0	0

53 = 16 numbers or ordered numbers + 9 yes/no items + 24 yes/no columns from the label options + 4 “was missing” flags. The logistic regression version leaves out one option for each of the 5 labels (53 − 5 = 48).

metadata flags for repeated people, the primary new-household cohort and the secondary new-person cohort. These flags are

TABLE VI
NO INFORMATION LOST: REBUILDING THE REMOVED FEATURES

Removed	Rebuilt from kept features	Children checked	Children that differ
School grade	age − years over-age − 5	1,307	0
Lost a parent	Three-valued logic from parent-alive items; unknown remains unknown	1,307	0
People per room	household size / rooms	1,307	0

All training children, including equal missing values; 3,921 checks in total.

TABLE VII
QUICK END-TO-END CHECK (5 DISTRICT-GROUPED FOLDS, X_TRAIN)

Model (untuned)	PR-AUC	ROC-AUC
Logistic regression (48 columns)	0.266 (±0.072)	0.627
XGBoost (53 columns)	0.252 (±0.089)	0.602
No skill (dropout rate)	0.144	0.500

Mean (± spread) over the 5 checking parts. Both models are untuned; this checks execution and exploratory training discrimination, not confirmed transport performance.

not model features.

VII. MODEL EVALUATION AND PROFESSOR-REQUESTED DIAGNOSTICS

A. *Training-only model comparison and calibration*

XGBoost remains the primary benchmark; LightGBM, CatBoost, ElasticNet/L1 logistic regression, Random Forest, and Extra Trees are compared on the same five district-grouped folds. All learned preprocessing, grade-KNN selection and class weights are fitted inside the appropriate training partition. Small, explicitly recorded hyperparameter grids are searched in inner grouped folds. Sigmoid and isotonic calibrators use training out-of-fold predictions with further fit/selection separation; validation and later outcomes do not fit these mappings. Raw and calibrated outputs are both retained because class weighting changes the meaning of uncalibrated probabilities.

The discrimination metric called PR-AUC in earlier sections is specifically scikit-learn average precision (AP), not trapezoidal integration. Recall and precision are reported at the same resource capacity, alongside Brier score, ten-bin equal-width calibration error (ECE), reliability curves, and confusion counts. Five-fold mean AP is distinct from AP of pooled out-of-fold predictions. The training-CV sigmoid mean AP selected Extra Trees (0.303), ahead of Random Forest (0.294) and XGBoost (0.259). This choice was frozen before the reported temporal run. The grids are limited and the training folds have been explored repeatedly; neither exhaustive optimization nor independent confirmation is claimed.

B. *Frozen retrospective temporal evaluation*

The primary evaluation contains 731 children from households absent in training, including 76 observed dropouts. The secondary evaluation contains 1,175 children absent as people from training, including 115 dropouts, but may share households. The 1,395-row later table is not wholly independent of training; district overlap also remains. Cohorts are defined by identifiers, not outcomes. Training-only regional prevalence defines high-risk regions (8 and 10); the definition is not updated using later labels.

The frozen policy ranks calibrated risk and selects the top 20% (146 primary or 235 secondary places), with deterministic hashed-identity tie breaking. Primary results are in Table VIII. All six families lose AP relative to their training CV. CatBoost and Random Forest perform differently in the later cohort, but these results do not authorize replacing the selected Extra Trees model after observing test outcomes.

Extra Trees primary recall is 26.3% and precision 13.7%; secondary recall is 24.3% and precision 11.9%. The primary fixed-prediction district-bootstrap 95% interval is approximately 0.103–0.182 for AP and 12.5–41.9% for recall (1,000 replicates). Intervals reflect clustered sampling of the observed cohort, not refitting, selection uncertainty, or future distribution changes. At 50% capacity the primary model catches 51/76 events but produces 314 false alarms. Increasing capacity changes the trade-off rather than solving it.

C. *Calibration and subgroup allocation*

Extra Trees primary mean calibrated risk is 12.4%, versus 10.4% observed dropout; Brier score is 0.0938 and ECE 0.0197. Secondary mean risk is 12.4%, versus 9.8% observed; Brier is 0.0900 and ECE 0.0263. A lower Brier score across periods can reflect lower prevalence, so it cannot alone establish improved calibration. Small high-risk bins have very few observations. Average calibration also hides regional errors and does not establish good ranking.

Regions 8 and 10 together receive 128 of the 146 primary places and contribute 17 true positives and 111 false alarms. Other regions contain 59 dropouts but receive only 18 places, catching three. In region 8, observed dropout falls from 28.0% in training to 12.8% (16/125) later, while mean predicted risk is 21.0%. Region 10 has only 20 later children and one event; its apparently large overprediction is sparse descriptive evidence. Under the same primary policy, both-parent households have 17/48 events caught (35.4%), compared with 3/20 (15.0%) in mother-only households. These are descriptive allocation disparities, not causal effects of household type. Counts and sparse-group flags accompany all subgroup results.

D. *Attrition, ablations, and uncertainty about mechanisms*

The attrition audit compares baseline covariates and parent co-residence among children with and without an observable subsequent outcome. Unknown outcomes are not relabelled as non-dropout. All predictive results therefore concern observable children; selection into follow-up limits population claims. A separate follow-up-weighted sensitivity is now documented in the controlled robustness extension; its assumptions and limitations remain explicit.

Training-only ablations remove missingness flags, geography, or the region predictor separately. Effects differ across models; removing all geography reduces mean AP in the compared

TABLE VIII
FROZEN TEMPORAL RESULTS WITH SIGMOID CALIBRATION AND TOP-20% ALLOCATION

Model	Primary: 731 children, 76 events AP	TP	FP	FN	Secondary: 1,175 children, 115 events AP	TP	FP	FN
XGBoost	0.147	21	125	55	0.132	34	201	81
LightGBM	0.135	22	124	54	0.125	33	202	82
CatBoost	0.154	22	124	54	0.139	29	206	86
ElasticNet/L1	0.151	21	125	55	0.136	32	203	83
Random Forest	0.151	23	123	53	0.133	34	201	81
Extra Trees	0.134	20	126	56	0.118	28	207	87
Age/sex comparator	0.083	10	136	66	0.081	18	217	97

TP: dropouts caught; FP: false alarms; FN: missed dropouts. Every model receives the same number of places. A random allocation expects 15.2 primary true positives at 146 places; it is an analytic expectation, not an independently fitted model. Extra Trees was selected on training validation, not this table.

TABLE IX
PRECISION, RECALL AND F1 AT THE SAME TOP-20% INTERVENTION CAPACITY

Model	Primary: 731 children Precision	Recall	F1	Secondary: 1,175 children Precision	Recall	F1
XGBoost	0.144	0.276	0.189	0.145	0.296	0.194
LightGBM	0.151	0.289	0.198	0.140	0.287	0.189
CatBoost	0.151	0.289	0.198	0.123	0.252	0.166
ElasticNet/L1	0.144	0.276	0.189	0.136	0.278	0.183
Random Forest	0.158	0.303	0.207	0.145	0.296	0.194
Extra Trees	0.137	0.263	0.180	0.119	0.243	0.160

Precision=TP/(TP+FP); recall=TP/(TP+FN); F1=2(precision × recall)/(precision+recall). Values are proportions (multiply by 100 for percentages), computed from the matching sigmoid-calibrated top-20% predictions in Table VIII.

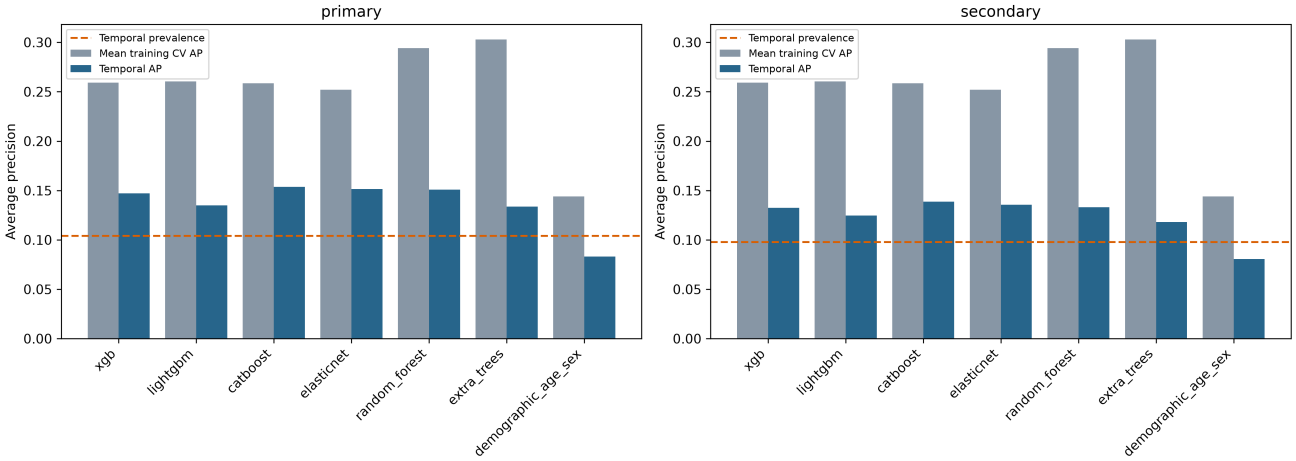


Fig. 9. Training cross-validation versus frozen retrospective temporal performance. The cohort and prevalence change; these differences describe transport failure rather than its cause.

configurations. This is sensitivity evidence, not proof that flags encode survey artifacts or that geographic predictors generalize. The temporal diagnosis does not justify retrospectively choosing a new ablation from test performance.

E. Global and local explanations

Held-out XGBoost explanations use native TreeSHAP on the raw log-odds scale. Penalized, class-weighted logistic coefficients are presented as model-score odds ratios, not population causal odds ratios. The final frozen Extra Trees fit is reconstructed from its saved training settings and required to reproduce temporal calibrated predictions (maximum error below 2×10^{-16}). Exact TreeSHAP [12] uses training tree-path counts as its reference. For this classifier, base value plus contributions reconstructs the *uncalibrated positive-class probability*, verified within 4×10^{-15}; the contributions do not sum to calibrated risk.

Signed one-hot contributions are summed within their source feature, then absolute values averaged for global ranking. This preserves local additivity but is not a new grouped-feature

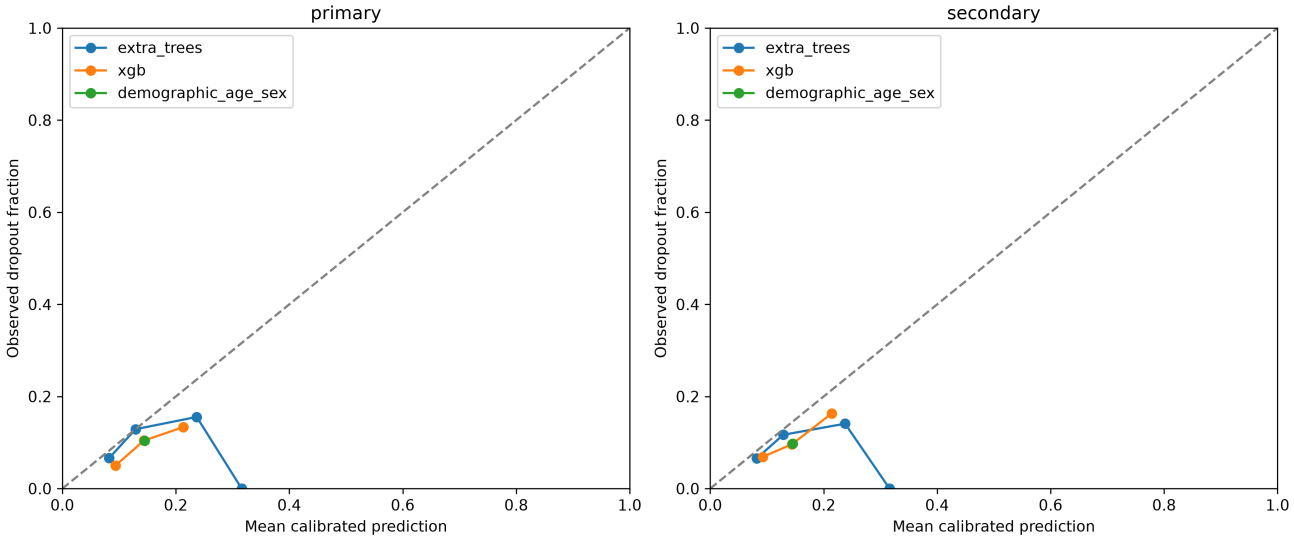


Fig. 10. Temporal calibration diagnostics using the frozen training-only mappings. No calibrator is refitted on these outcomes.

Shapley game. Region is the largest primary contribution (mean absolute 0.0759), followed by teacher-absence missingness (0.0320) and maternal education (0.0180). One example per TP/FP/FN/TN category in each cohort is selected by the smallest pseudonymous identity hash, avoiding attractive-example selection. SHAP explains model associations; it cannot establish that changing a feature will reduce dropout.

F. Source-audited cleaning and exploratory alternatives

The original raw/processed tables and temporal results are retained. Separate corrections recognize Wave 1 vacation code 99 in either hours or minutes, mark a conflicting duplicate English-score record unknown, and treat a household asset total as missing when an owned item has unknown/negative value. These corrections preserve row identities, targets, and the target-last schema. Further alternatives use routed subject-teacher absence and relax attendance assumptions. Teacher absence is a proxy, not total lost instruction days; unknown later holiday status is not filled by an inferred holiday. Ambiguous spending blanks and unsupported new plausibility cut-offs are not reinterpreted.

On identical training folds with fixed baseline settings and training-only sigmoid calibration, core corrections change Random Forest mean AP from 0.287 to 0.298 and true positives from 75 to 78 at 260 places. Extra Trees changes from 77 to 79 true positives, while its mean AP falls slightly from 0.286 to 0.285. CatBoost worsens. The routed-attendance Random Forest sensitivity catches 80 events with 180 false alarms, but its score is not directly comparable with the earlier nested-tuned selection. These are mixed, exploratory gains; no improved temporal performance has been established. Definite recording corrections should not be undone to obtain a higher score, and uncertain assumptions should not be adopted solely for a CV gain.

Raw missed-hours missingness changes from 77.8% in training to 13.5% in the primary later cohort, and teacher-absence missingness from 56.8% to 31.7%. Among observed missed-hours values, medians are 3 and 0 and KS distance is 0.492. These raw diagnostics precede the pipeline's blank-to-zero rule; they are not post-imputation model-input comparisons. Survey routing, composition and behavior can all contribute, so the shift does not identify a single cause of error.

Earlier policy experiments also remain separate. A high-recall training policy catches 185/188 events but flags 1,284/1,307 children. Literacy, funding and routed-teacher enrichment and F1/F2-oriented alternatives expose similar false-alarm trade-offs. They do not establish a deployable policy or supersede the frozen capacity comparison. The demographic age/sex comparator and random allocation provide interpretable resource-matched references.

VIII. Controlled Robustness Extension

Purpose and preservation. This separately specified extension (28 October 2026) addresses calibrated ablations, temporal geography comparisons and follow-up weighting. The frozen baseline above is retained. Source-confirmed vacation-code, conflicting-assessment and incomplete-asset corrections use separate tables with identical children and outcomes; the target stays last. Uncertain attendance alternatives are not silently adopted.

Candidate comparison. Six families are crossed with four configurations (full, no explicit missingness flags, no geography, neither) and three class-weight strengths (none, square-root imbalance ratio, full ratio): 72 fixed candidates. Geography removal also removes geography from KNN matching. Model complexity is held at the established baseline settings; this bounded family/feature/weight search is not unrestricted hyperparameter tuning. Removing explicit flags does not remove

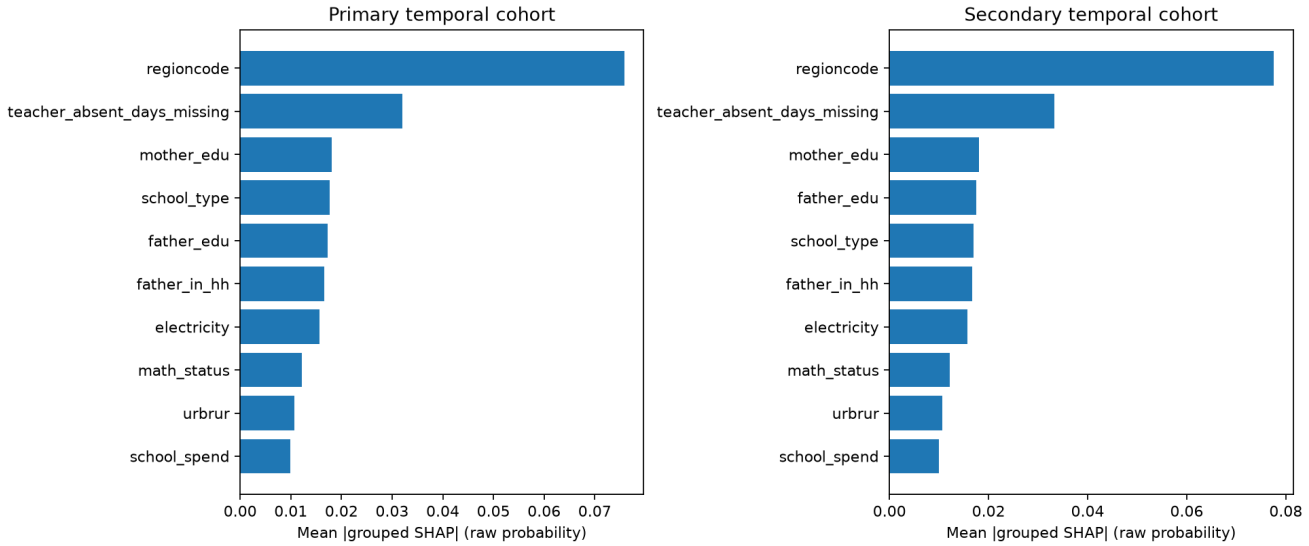


Fig. 11. Frozen Extra Trees global explanations, grouped by original feature. Contributions are on the raw probability scale, not calibrated-risk differences.

Reading Figure 11: what drives the predictions. The bars summarize the average magnitude of each feature's contribution; they do not show whether that feature raises or lowers every child's score. Region has the largest contribution in the primary cohort (0.0759), followed by teacher-absence missingness (0.0320) and maternal education (0.0180). A large missingness contribution means the model uses whether information was recorded, as well as the recorded values. Because recording patterns and regional dropout rates change between survey waves, these associations require stability checks before operational use. They are not evidence that region or missingness causes dropout.

Connecting the global pattern to Table X. The four rows illustrate a missed dropout (FN), a false alarm (FP), a correctly unflagged child (TN), and a correctly flagged dropout (TP). They are reproducibly selected examples, not typical cases or estimates of each group's performance. Positive contributions raise the raw score relative to the model's base value; negative contributions lower it. Only the three largest absolute contributions are displayed, so these entries alone cannot reconstruct the full score. The complete contributions sum with the base value to the raw probability, and a separate training-fitted calibration mapping produces the reported risk.

Practical implication. The primary policy ranks children and flags 146 of 731 at the fixed 20% capacity; it does not use a 50% probability threshold. A child with calibrated risk below 50% can therefore be flagged. Across the full cohort, this catches 20 of 76 observed dropouts with 126 false alarms. The examples help explain individual scores, but do not establish that the intervention policy is accurate or that changing a highlighted feature would prevent dropout.

TABLE X
DETERMINISTIC LOCAL EXTRA TREES EXAMPLES FROM THE PRIMARY TEMPORAL COHORT

Outcome	Hash prefix	Raw score	Calibrated risk	Three largest absolute grouped SHAP contributions
FN	02e0e988	0.251	0.081	regioncode (-0.062); teacher_absent_days_missing (-0.052); mother_edu (-0.030)
FP	02773018	0.480	0.160	regioncode (-0.046); teacher_absent_days_missing (-0.041); father_edu (+0.022)
TN	014afcbd	0.252	0.082	teacher_absent_days_missing (-0.057); father_edu (-0.040); school_type (-0.036)
TP	0a79d5c4	0.524	0.179	father_in_hh (-0.032); father_edu (+0.025); regioncode (-0.020)

Smallest full identity hash in each outcome class. Signs show contribution relative to the raw-score base value, not causal changes or calibrated-risk increments. All contributions, the base value, and full pseudonymous hashes are exported in the diagnostic tables.

all missingness information from survey-status categories or imputed values.

Selection and calibration. Within each of five outer district-grouped folds, capacity selection maximises pooled recall under fold-specific top-20% allocations in three inner folds; mean fold AP and fixed candidate order break ties. The AP control reverses the first two criteria. Three calibration folds repeat the entire selection within their own training partitions before fitting a sigmoid mapping. No outer or later outcome selects a setting or fits a calibrator. The 24 balanced-weight family/feature ablations use fixed estimators with training-only OOF calibration. A separately declared equal-weight mean of the six full-feature calibrated models provides an ensemble comparison; averaging does not guarantee ensemble calibration.

Capacity selection catches 20 of 76 events (recall 26.3%), with 126 false alarms (precision 13.7%). The paired recall difference from the frozen baseline is +0.0 percentage points; its 95% district-bootstrap interval is [-3.5, +3.6]. These intervals condition on fitted models and fixed selected sets, not all development uncertainty. A small point improvement is not

TABLE XI
EXPLORATORY ROBUSTNESS RESULTS: PRIMARY COHORT, 731 CHILDREN, 76 DROPOUTS AND 146 INTERVENTION PLACES

Procedure	AP	TP	FP	FN
Frozen tuned Extra Trees	0.134	20	126	56
Capacity-selected	0.136	20	126	56
AP-selected control	0.130	21	125	55
Equal-weight ensemble	0.141	22	124	54
Follow-up-weighted Extra Trees	0.135	21	125	55
Region-stress capacity	0.158	23	123	53
Region-stress AP	0.158	23	123	53

New procedures use the same corrected-data cohort and capacity. Later outcomes do not select a replacement for the frozen model.

proof of better future performance.

The five outer folds yield 73 true positives and 187 false alarms for capacity selection, versus 67 and 193 for the AP control, at 260 total places. The historical tuned Extra Trees result was 76 and 184, but its family was selected using the earlier outer-fold comparison; the new procedure evaluates family selection inside the folds as well. These differently selected CV summaries are not a clean estimate of a performance gain.

Follow-up sensitivity. The baseline has 2,532 children: 1,374 with known follow-up (including 67 basic-education completers) and 1,158 unknown. Completers are observed, not attrition. A baseline-predictor logistic observation model with grouped cross-fitting obtains AUC 0.614 and Brier score 0.239. Uncapped stabilised weights among observed children range from 0.624 to 2.942, with effective sample size 1,265; caps of 5 and 10 do not change this full-baseline diagnostic. Stability does not establish ignorable missingness.

A separate fixed Extra Trees sensitivity estimates observation weights within each allowed training partition, excluding validation districts. Stabilised weights capped at 5 enter model fitting and OOF sigmoid calibration. The target remains dropout conditional on non-completion. Interpretation requires the full follow-up status (including completion) to be missing at random given baseline predictors, positivity and adequate observation-model specification. These assumptions cannot be verified from unknown outcomes. Reported held-out performance remains unweighted among observed children; it is not measured performance among lost children.

Regional stress test. The adaptive regional stress test retains the candidate grid and outer district folds, but inner selection holds out complete regions. Calibration repeats this selection inside its training subsets. Capacity selection catches 23 primary dropouts with 123 false alarms (AP 0.158); AP selection catches 23 with 123 false alarms (AP 0.158). Both final regional procedures select Random Forest without geography and with balanced class weights. Their outer-validation true positives are 61 (capacity) and 65 (AP), versus 73 and 67 under district-based selection, so regional stress does not improve internal recall. The regional capacity procedure's primary paired recall difference is +3.9 percentage points, with a 95% interval [-11.4, +19.2]; this includes no improvement. This hypothesis arose from preceding retrospective failures. Although later outcomes never enter fitting or selection, these results are exploratory and cannot be presented as independent confirmation.

IX. LIMITATIONS AND REMAINING WORK

The model misses most primary temporal dropouts at the specified 20% capacity and produces many false alarms. It should not be presented as a validated operational early-warning system. Regional instability, measurement changes, attrition, limited event counts and repeated training-data experimentation limit the conclusions. Repeating the same temporal evaluation verifies reproducibility; it does not create a new independent test.

The calibrated feature ablations, temporal no-geography evaluation and follow-up-weighted sensitivity are now implemented as exploratory extensions. Completing them does not establish accurate prediction or eliminate missing-outcome bias. Future feature, geography, calibration or policy changes need a separately specified evaluation and an independent future/external cohort. The evidence supports a reproducible research pipeline, not a deployment-ready classifier.

X. EXECUTION VERIFICATION AND EVIDENCE FILES

The original 41-code-cell notebook stage executed from a fresh kernel without error and passed 83 tests. Robustness and weighting experiments were then executed separately; three appended evidence-display cells were checked in a fresh kernel, and all 100 current tests passed. The combined notebook was not rerun from scratch in this extension. Historical dependency checks found no broken requirements. The regenerated outputs were compared with the prior saved results. See `phase2_implementation/final_validation.json` for original integration checks, and `phase2_implementation/robustness_validation.json` for extension checks and hashes.

The notebook produces the original eight figures and the added model diagnostics. Prefixes `calibration_`, `subgroup_`, `demographic_`, `interpretation_`, `temporal_`, and `recleaning_` identify the associated evidence tables under `phase2_implementation/`. The frozen protocol retains its historical pre-run wording; execution manifests and this report record completed work. The 32-entry decision log is synchronised by the appended extension cells

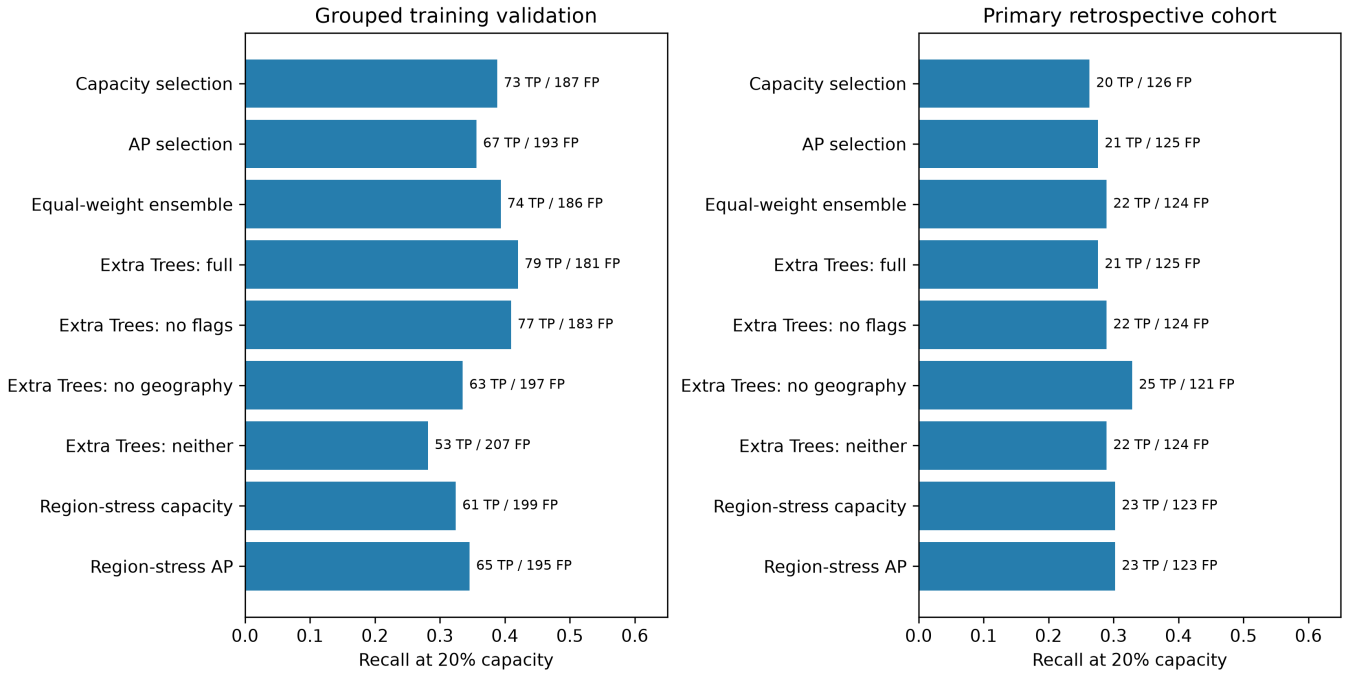


Fig. 12. Controlled robustness at the same intervention capacity. Training results are grouped outer-fold predictions; later results are exploratory retrospective evidence.

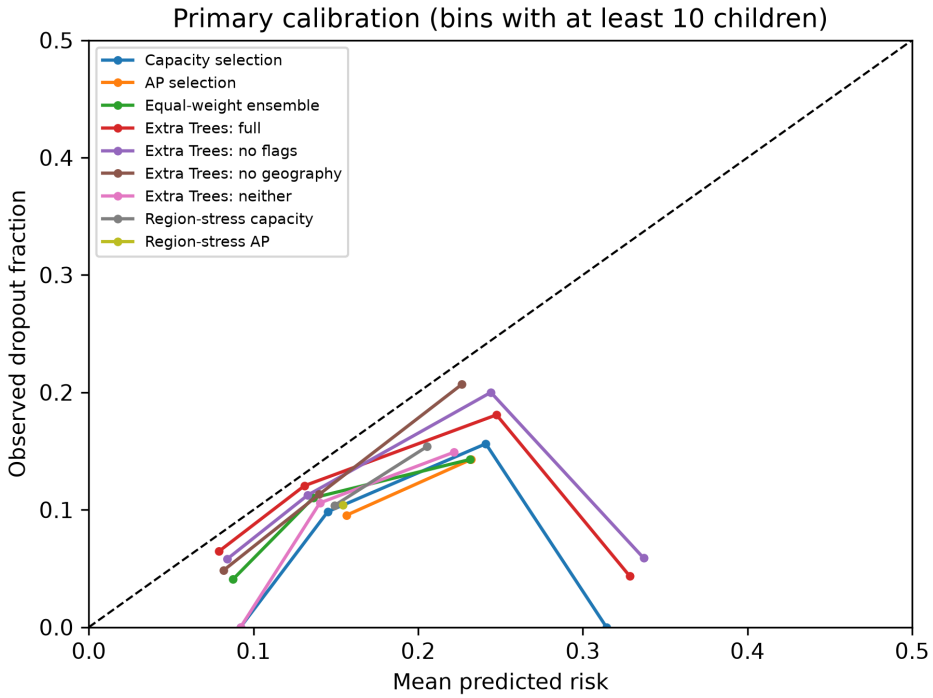


Fig. 13. Primary-cohort calibration for revised procedures and Extra Trees ablations. Bins with fewer than ten children are omitted here but retained in the exported tables.

so a rerun cannot silently revert it to the earlier “temporal pending” status. Report source and PDF remain together in the required Phase 2 directory.

APPENDIX: PLAIN-LANGUAGE GUIDE TO THE TECHNICAL TERMS

The terms below are used in this report; terms already explained in the Phase 1 report keep the same meaning.

Pipeline: A chain of preparation steps followed by a model, run in a fixed order. Fitting it learns every setting from the training data at once.

ColumnTransformer: The part of the pipeline that applies the right preparation to each group of columns side by side (for example, a middle-value fill for lopsided numbers and one-hot encoding for labels).

Fit / transform: *Fit* = learn the settings (e.g. a middle value or a cap limit) from data; *transform* = apply the stored settings to any data without learning anything new.

X_train / X_test: The children used to build the model (2009/10) and the children kept aside to test it (2013/14).

Leakage: When information from the data used for checking sneaks into building the model, making results look better than they really are.

Fold, building part, checking part: Cross-validation splits X_train into 5 parts; the model is built on 4 (the building part) and checked on the 5th (the checking part), rotating 5 times.

Custom transformer: A preparation step written for this project (in `src/transformers.py`) that follows the same fit / transform rules as the standard scikit-learn steps.

KNN (nearest neighbours) filling: Filling a missing grade from similar training children with numerical differences and category matches treated separately; neighbour count selected within training.

Capping: Pulling an extreme value down to a set limit instead of deleting the child.

Rank (0–1): The position of a value among the training values: 0 = lowest, 0.5 = middle, 1 = highest.

Consumer price index, inflation factor: The official measure of how prices rise; dividing 2013/14 amounts by 1.62 expresses them in 2009/10 prices.

One-hot encoding: Turning a label with several options (e.g. 10 regions) into separate yes/no columns.

Standardising: Putting numbers on a common scale with average 0 and spread 1.

Correlation (r, ρ): How closely two features move together, from -1 to $+1$. The ordinary (Pearson) r only sees straight-line relationships; the rank (Spearman) ρ also sees curved ones.

VIF: How much a feature is already explained by all the others together (above 10 = redundant).

Mutual information: How much knowing a feature reduces uncertainty about dropout; it captures any kind of relationship.

PCA, component, scree plot: PCA combines many columns into a few new ones (components) that keep as much of the variation as possible. A scree plot shows how much variation each component keeps, and the cumulative line how much the first k components keep together.

Class weighting: Counting each dropout several times more during training so the model pays enough attention to the rare group.

PR-AUC, ROC-AUC: Scores from 0 to 1 for how well a model puts real dropouts at the top of its risk list. The AP reference is the evaluated cohort prevalence (0.144 in training, 0.104 in the primary later cohort); the ROC-AUC reference is 0.5.

Calibration: Adjusting risk scores toward observed event frequencies using training-only held-out predictions. Calibration can fail under a later distribution shift.

Tie-break rule: The Phase 1 rule used when two repeating features carry similar information: keep the one measured directly and remove the one calculated from others.

REFERENCES

[1] E. L. Muhigira, "ElimuSentry - Phase 1: Preparing the data," 04-652 Artificial Intelligence System Design, Assignment 2, Carnegie Mellon University Africa, 2026. [Online]. Available: https://github.com/Emile-Lucky-Muhigira/AISD_Project

[2] Institute of Statistical, Social and Economic Research (University of Ghana), Economic Growth Center (Yale University) and Northwestern University, *Ghana Socioeconomic Panel Survey, Waves 1–3* [Dataset], Harvard Dataverse, 2022, doi:10.7910/DVN/E5QP0F.

[3] World Bank, "Inflation, consumer prices (annual %), Ghana," World Development Indicators, series FP.CPI.TOTL.ZG. [Online]. Available: https://fred.stlouisfed.org/series/FPCPITOTLZGGHA

[4] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[5] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD*, 2016, pp. 785–794.

[6] S. Kaufman, S. Rosset, C. Perlich and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, 2012.

[7] I. T. Jolliffe and J. Cadima, "Principal component analysis: A review and recent developments," *Philosophical Transactions of the Royal Society A*, vol. 374, 20150202, 2016.

[8] J. W. Tukey, *Exploratory Data Analysis*. Reading, MA: Addison-Wesley, 1977.

[9] A. Kraskov, H. Stögbauer and P. Grassberger, "Estimating mutual information," *Physical Review E*, vol. 69, 066138, 2004.

[10] D. Filmer and L. H. Pritchett, "Estimating wealth effects without expenditure data—or tears: An application to educational enrollments in states of India," *Demography*, vol. 38, no. 1, pp. 115–132, 2001.

[11] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, e0118432, 2015.

[12] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4765–4774.